

## Education

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### Cornell University

*Doctor of Philosophy in Robotics (Mechanical)*

Advisor: Dr. Robert Shepherd (Organic Robotics Lab)

Ithaca, NY

**Expected Graduation: May 2030**

### William Marsh Rice University

*Bachelor of Science in Mechanical Engineering*

Minor in Engineering Design

Advisor: Dr. Kaiyu Hang (RobotPI Lab)

Houston, TX

**Graduation: May 2025**

Distinction in Research and Creative Works

## Research Interests

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Soft robotics; proprioception and state estimation; biomechanics and biomimicry; volumetric additive manufacturing; in-hand manipulation; mechatronic system design

## Publications

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**2025**

C. Darkes-Burkey, Y. Park, G. Eken, **H. Webb**, R. Shepherd, Controlling Convection in Volumetric Additive Manufacturing for Large Volume Structures at Extreme Throughput, *Advanced Functional Materials* 2025

**H. Webb**, P. Changrunmaneeikul, S. Yuan, and K. Hang; Wearable Roller Rings to Augment In-Hand Manipulation through Active Surfaces, *IEEE International Conference on Intelligent Robots and Systems (IROS)* 2025

## Patents

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**2024**

**H. Webb**, S. Yuan, and K. Hang; “Wearable roller rings to enable robot dexterous in-hand manipulation through active surfaces” - US Patent App. 19/256,848

## Presentations & Posters

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**2026**

**H. Webb**, Y. Park, R. Shepherd, 6-DoF Fiber Optic Strain Sensor via Geometric Variable Strain Model and Machine Learning for Robust Soft Robot Proprioception *Sibley School for Mechanical and Aerospace Engineering (MAE) Reunion 2026* [Poster]

**2025**

**H. Webb**, P. Changrunmaneeikul, S. Yuan, and K. Hang; Wearable Roller Rings to Augment In-Hand Manipulation through Active Surfaces. *IEEE International Conference on Intelligent Robots and Systems (IROS)* 2025 [Talk/Poster]

K. Noonan, B. Ramsey, J. Villarreal, **H. Webb**; Automated Screen Printer for Fabrication of Electrodes for Biosensors. Rice University Huff Engineering Design Showcase [Poster]

**2024**

**H. Webb**, P. Changrunmaneeikul, S. Yuan, and K. Hang; Wearable Roller Rings to Enable Robot Dexterous In-Hand Manipulation through Active Surfaces. ASME Mechanism and Robotics Committee [Talk/Poster]

- H. Webb**, P. Changrunmaneeikul, S. Yuan, and K. Hang; Wearable Roller Rings to Enable Robot Dexterous In-Hand Manipulation through Active Surfaces. Texas Regional Robotics Symposium (TEROS) [Poster] **2022**
- J. Chang, R. Lee, **H. Webb**; Body Cooling Device for the Cooling of Humans in Heat Waves. Rice University Engineering Design Showcase [Poster]

### **Honors & Awards**

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- Sigma Xi Membership Nomination** **Oct. 2025**  
*Sigma Xi Scientific Research Honor Society*
- Graduate School Dean's Scholar Engagement Seminar** **Sep. 2025**  
*Cornell University*
- IROS-SDC (Student and Developing Countries) Travel Award** **Aug. 2025**  
*IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*
- Distinction in Research and Creative Works** **May 2025**  
*William Marsh Rice University*
- Undergraduate Teaching Fellow, Mechanical Engineering** **Jan. 2025**  
*William Marsh Rice University*
- 3rd Place, Student Mechanism and Robot Design Competition (SMRDC)** **Aug. 2024**  
*American Society of Mechanical Engineers (ASME)*
- ASME Student Financial Support Award** **Jun. 2024**  
*American Society of Mechanical Engineers (ASME)*
- Undergraduate Teaching Fellow, Mechanical Engineering** **Jan. 2024**  
*William Marsh Rice University*

### **Teaching Experience**

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- Teaching Assistant (MAE 4630 - Advanced Product Design)** **Jan. 2026 - May 2026**  
*Cornell University, Mechanical and Aerospace Department*
- Led independent instruction sessions, aided with group projects, and assisted with grading and material for 60+ graduate and undergraduate students
- Head Teaching Assistant (ENGI 210 - Prototyping & Fabrication)** **Jan. 2023 - May 2025**  
*Rice Makerspace, Oshman Engineering Design Kitchen (OEDK)*
- Taught material, hosted tutorials, and led workshops on engineering tools and fabrication methods
  - Assisted 50+ graduate and undergraduate students with team projects and individual competencies in engineering design
- Teaching Assistant (MECH 310 - Rigid Body Dynamics)** **Jan. 2025 - May 2025**  
*Rice Undergraduate Teaching Fellow Program, Mechanical Engineering Department*
- Assisted with teaching and grading of material for 50+ undergraduate students in an upper level course

**Teaching Assistant (MECH 310 - Rigid Body Dynamics)**

*Jan. 2024 - May 2024*

*Rice Undergraduate Teaching Fellow Program, Mechanical Engineering Department*

- Assisted with teaching and grading of material for 50+ undergraduate students in an upper level course

**IBB Summer Academy Assistant**

*Jun. 2022 - Jul. 2022*

*Rice Department of Innovation*

- Led week-long courses for high school students on engineering design tools, devices, and processes

**Research & Technical Experience**

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**Lead Engineer, Team Pressing Matters**

*Aug. 2024 - May 2025*

*Senior Capstone Design Project*

- Oversaw team deadlines and communication with sponsors and collaborators, ensuring proper communication on work between the client and development team throughout the project
- Performed several technical design reviews on mechanical subsections to ensure feasibility and design optimization in terms of modularity, cost, and ease of construction

**Co-President**

*May 2023 - May 2024*

*Rice Robotics Club*

- Managed and aided 10+ subteams and 50+ students on the development of various robotics projects such as several featherweight combat robots and a biomimetic four-legged walking robot
- Developed new safety and testing procedures for R&D and Combat Robotics subteams
- Organized and instructed several workshops, talks, and presentations on robotics and engineering skills

**Team Lead, Mirrored Robot Arm Subteam**

*May 2022 - May 2023*

*Rice Robotics Club*

- Led the development and construction of 6-DoF underactuated robot arm and 17 DoF robot hand
- Promoted collaboration on the integration of manufactured components with electronics subteams to allow for primitive mirroring of human movement in a small-form device

**Team Lead, Highschool Aerospace Scholar**

*Oct. 2019 - Jul. 2020*

*National Aeronautics and Space Administration (NASA)*

- Led 20+ students on the design of a theoretical human landing system between Gateway Space Station and the lunar surface utilizing integrated CAD environment
- Collaborated with other teams to integrate Lunar Lander into theoretically designed Gateway Space Station for mock Artemis Mission

**Mentorship & Service**

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**First Year Liaison**

*May 2026 - Present*

*Sibley Graduates in Mechanical and Aerospace (SiGMA), Cornell University*

- Facilitated department engagement among new students, organized inter-year social events, and coordinated mock first year Qualifying Exams

**Robotics Club Mentor**

*May. 2025 - Present*

*Conroe High School, Academy for Science and Health Professions (ASHP)*

- Lead discussion and assisted high school students with designing, controlling, and building robots for their regional FRC competition (Texas District)

**AP Research Expert Advisor**

*Aug. 2025 - May 2026*

*Conroe High School, Academy for Science and Health Professions (ASHP)*

- Lead student in design for an automated object detection for individuals using mobility assistance devices

**FIRST Robotics Competition (FRC) Volunteer**

*Aug. 2024- May 2025*

*FRC Texas District*

- Aided local competitions in coordinating team movements as well as judging and running game rounds

**Selected Projects**

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**Automated Screen Printer for Fabrication of Electrodes for Biosensors**

*Aug. 2024 - May 2025*

*Senior Capstone Design Project*

- Fully developed an automated screen printer for electrode fabrication in the Sempionatto Research Group capable of taking in an electrode vector file and outputting fully screened sensors with a 95% yield
- Independently designed and constructed a working gantry for movement in all cartesian directions
  - Device's machined parts were fabricated through the use of various subtractive manufacturing methods such as plasma-cutting, waterjet-cutting, and laser-cutting
- Designed a primitive weighted motion planner for ink and screening toolpaths, generating task-space coordinates with motor control flags in G-code format for electronics bus
- Created closed-loop velocity and position control for use in gcode execution and device homing/jogging
- Developed a custom UI for preprocessing electrode vector files, modeled after common 3D printing slicers to streamline tool path generation, repeated electrode Cricut file, and parameter tuning

**In-Hand Manipulator for Parallel Jaw Grippers**

*Aug. 2024 - May 2025*

*Independent Undergraduate Research*

- Designed a robot hand set for parallel jaw grippers to provide full spatial manipulation through the use of active surfaces, deformable fin-rays, and a compliant double blade rotary beam system
- Developed a modular, parameterized backing for quick installation on robots of all levels of fidelity in order to upgrade their manipulation capabilities

**Electronic Prony Brake Torquemeter**

*Jun. 2024 - Aug. 2024*

*Rice Mechanical Engineering Department*

- Designed and developed multiple functional 3D-Printed Prony Brakes to mechanically test the concepts of dynamic torque, speed, and power through inset motors and load cells.
- Built an electronic subsystem consisting of a photo interrupter, Arduino microcontroller, and signal amplifier to calculate the power output of a motor for use in an upper level mechanical engineering course.

**Wearable Roller Rings to Augment In-Hand Manipulation through Active Surfaces**

*Aug. 2023 - May 2024*

- Designed in-hand manipulator for both human & robotic grasping systems for differential, non-holonomic, and full spatial manipulation through angled active surfaces
- Created compliant attachment using inverted quatrefoil geometry for zero-cost installation on graspers
- Developed a generalized active surface-based motion model for arbitrary geometry objects

**Technical Skills**

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**Software:** SolidWorks, Onshape, Eagle, ROS

**Programming Languages:** Python, C++, Matlab

**Technical:** Product Manufacturing (3D Printing, CNC, Lathe), Embedded System Design, Welding (MIG, Stick)

**Tools:** Microsoft Office (Word, Excel, Powerpoint), LaTeX